Vanessa Sochat

vsoch.github.io

PROFESSIONAL SUMMARY

Software engineer that learns quickly, and works with passion and urgency. Expertise includes:

- Converged computing and Kubernetes
- Container technologies and registries
- Next-generation graph scheduling
- Orchestration of scientific workflows
- Agentic collaboration
- Machine Learning
- Web interfaces and APIs
- Binary Analysis

EXPERIENCE

Principal Computer Scientist, Livermore Computing: *Lawrence Livermore National Laboratory, CA*

February 2021 – present

- Designed and implemented agentic system for build, deploy, and HPC workload optimization
- Expertise in converged computing, batch orchestration between cloud and HPC
- Leading project to implement production user-space Kubernetes on-premises
- Lead developer of the Flux Operator for Kubernetes and the Flux RESTFul API
- Leading community outreach for Flux, tutorials for HPCKP25, ISC25, HPCIC25, and SC25
- Collaborations with Google Cloud, AWS, IBM/RedHat, and Azure

Software Engineer, Research Computing: *Stanford University, Stanford CA*

Sept 2016 – January 2021

- Systems engineer and architecture, infrastructure and standards development, and specialized technical consultation to better ensure reproducibility of scientific computational analyses and workflows
- Implemented Snakemake Google Life Sciences executor backend using several Google Cloud APIs
- Host and producer of Research Software Engineer Stories podcast
- Designed the Scientific Filesystem for organization and discovery of scientific applications
- Developer of Singularity Hub, Singularity Registry, and original developer for Singularity containers
- Lead of open source project The Experiment Factory for reproducible behavioral experimentation
- Implemented and serviced image processing pipelines for the School of Medicine.
- Complete list of work available at https://vsoch.github.io/work

PhD Candidate, Poldrack Lab: *Stanford University, Stanford CA*

June 2011 – Aug 2016

- Designed and developed a Dockerized infrastructure, expfactory.org, to deploy web-based experiments
- Conceptualized and implemented open source software, Wordfish, for generating custom NLP pipelines
- Created an interactive, reproducible workflow to for genomic, behavioral, and brain imaging analyses
- Identified optimal parameters for comparison of statistical brain maps using classification framework
- Imagined and created web viewers for brains using nodeJS, a neuroimaging data model, and FileReader
- Built clinical web application to explore anatomical and genomic features associated with brain tumors

- Built model and database to classify artifact in functional MRI using regularized logistic regression
- Created complicated analysis pipelines in a HPC environment to analyze thousands of brain images

Data Technician, Laboratory of Neurogenetics: *Duke University, Durham NC*

May 2009 – May 2011

- Coded and deployed image processing pipelines in HPC environment using python, bash, and Matlab
- Wrote custom tools to check the quality of brain images, organize data, and interact with participants
- Responsible for creating and administering a battery with over 30 cognitive paradigms using Qualtrics

Founder, Goggles Optional Podcast: *Stanford University, Stanford CA*

Nov 2013 – present

- Developed and currently maintain infrastructure for a weekly science podcast with over 50K downloads
- Weekly responsibility to generate episode content, update databases, and publish

Student Director, Informatics Concentration for MD Students: *Stanford CA*

May 2013 – May 2015

- Organized quarterly sessions for approximately 30 medical students interested in informatics
- Set up social media groups and advertising for MD student recruitment

Teaching Assistant, Biomedical Image Analysis and Interpretation: *Stanford CA*

Jan 2013 – May 2014

- Created new course content for 10 lectures, including interactive slides and class handouts
- Single handedly developed two new projects, including a database of "cookie tumor" images
- Taught weekly section meetings, and gave two full lectures on machine learning and neuroinformatics

SKILLS AND QUALIFICATIONS

Computer Experience

Languages:	Go, Python, shell, C/C++, Groovy, R, JavaScript, Matlab, Web
Databases:	MySQL, PostgreSQL, neo4j, couchdb, Big Query, sqlite3, cloud
Infrastructure:	Kubernetes, Docker, Singularity, VirtualBox, Vagrant
Visualization:	D3, canvas, Shiny (R), Photoshop, Illustrator, Maya, Blender

Data Analysis

High Performance Computing:	Flux Framework, MPI, Slurm, SGE, HTCondor
Data Structures	JSON, xml/RDF, yaml, hcl

Web Development

Frameworks:	FastAPI, Django, Jekyll, Flask, Wordpress, nginx, wasm, uWSGI
Cloud Technology	Google Cloud, AWS, Azure, IBM Cloud
Continuous Integration	GitHub Actions, CircleCI, Travis
Version Control	GitHub, GitLab, Bitbucket, Gogs

EDUCATION

PhD, Biomedical Informatics: *Stanford University, Stanford CA*

Sept 2011 – Aug 2016

- Stanford Graduate Student Fellowship (Albion Walter Fellow)
- Microsoft Graduate Women's Scholar (2012)
- National Science Foundation Graduate Fellowship

BA in Psychology and Neuroscience: *Duke University, Durham NC*

Aug 2005 – June 2009

- Magna Cum Laude, Dean's List, Member of Psi Chi, the National Honor Society in Psychology

IN PREPARATION / UNDER REVIEW

Sochat, V, and D. Milroy "Hierarchical Server Architecture for Agentic Science" *Anticipated submission, April 2026.*
Sochat, V, and D. Milroy "Agentic Orchestration of HPC Applications in Cloud" *Anticipated journal submission, May, 2026.*

PUBLICATIONS

Sochat, Vanessa and Daniel Milroy (2025). "Agentic Orchestration of HPC Applications: A case study with Google Gemini in Cloud." *White Paper*, November 2025.

Sochat, Vanessa and Daniel Milroy (2025). "Possible Futures for Cloud Cost Models." In *arXiv preprint arXiv:2511.01862*.

V. Sochat, L. Pottier and D. Milroy (2025). "State Machine Orchestration of an HPC Workflow in Cloud," in Proceedings of the SC'25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis, pp. 2293-2304.

V. Sochat, D. Milroy, A. Sarkar, A. Marathe and T. Patki (2025). "Usability evaluation of cloud for HPC applications," in *Proceedings of the SC'25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*, pp. 135-150.

Sochat, Vanessa and Daniel Milroy (2025). "If We Build Them, They Will Run: Automated HPC Apps Deployment and Profiling with eBPF in Cloud." In *F1000Research*, vol. 14, no. LLNL-JRNL-2012728.

Jolicoeur, Lise, **Sochat, Vanessa**, et al**.** "Enabling RDMA and GPUs in Rootless Kubernetes for Accelerated HPC and AI Applications." International Euro-Par Workshops, VHPC. 2025.

Sochat, Vanessa. 2025. “Trends for Pulling HPC Containers in Cloud.” In *CLOUD COMPUTING 2025, The Sixteenth International Conference on Cloud Computing, GRIDs, and Virtualization, 69–80.*

V. Sochat, D. Milroy, C. Misale, J. Luettgau, E. F. Bollig and W. Magro, "Converged Computing: A Best of Both Worlds of High-Performance Computing and Cloud," in *Computing in Science & Engineering*, vol. 26, no. 3, pp. 4-7, July-Sept. 2024, doi: 10.1109/MCSE.2024.3489732.

Sochat V, Culquicondor A, Ojea A and Milroy D. The Flux Operator [version 1; peer review]. F1000Research 2024, 13:203 (https://doi.org/10.12688/f1000research.147989.1)

Sochat, Vanessa, David Fox, and Daniel Milroy (2024). “A contention-free model for converged kubernetes on hpc.” In *arXiv preprint arXiv:2406.06995*.

Lancaster, Alexander K., Richard M. Single, Steven J. Mack, **Vanessa Sochat**, Michael P. Mariani, and Gordon D. Webster. 2024. “PyPop: A Mature Open-Source Software Pipeline for Population Genomics.” *Frontiers in Immunology* 15. https://doi.org/10.3389/fimmu.2024.1378512.

Gina Turco, Christie Chang, Rebecca Y. Wang, Griffin Kim, Emily H. Stoops, Brianna Richardson, **Vanessa Sochat**, Jennifer Rust, Rose Oughtred, ..., Anastasia Baryshnikova. May 2023. Global analysis of the yeast knockout phenome.Sci. Adv.9,eadg5702. DOI:10.1126/sciadv.adg5702

Sochat, Vanessa, Matthieu Muffato, Audrey Stott, Marco De La Pierre, and Georgia Stuart. 2023. “Automated Discovery of Container Executables” 11 (1): 6. https://doi.org/10.5334/jors.451

Sochat, V., May, N., Cosden, I., Martinez-Ortiz, C. and Bartholomew, S., 2022. The Research Software Encyclopedia: A Community Framework to Define Research Software. Journal of Open Research Software, 10(1), p.2. DOI: http://doi.org/10.5334/jors.359

Sochat et al., (2021)**.** Collaborative Container Modules with Singularity Registry HPC. Journal of Open Source Software, 6(63), 3311, https://doi.org/10.21105/joss.03311

Sochat, V. 2021. “The 10 Best Practices for Remote Software Engineering.” May 1, 2021. Communications of the ACM, May 2021, Vol. 64 No. 5, Pages 32-36. 10.1145/3459613 https://cacm.acm.org/opinion/articles/252174-the-10-best-practices-for-remote-software-engineering/fulltext.

D.S. Katz, M. Gruenpeter, T. Honeyman, L. Hwang, M.D. Wilkinson, **V. Sochat**, H. Anzt, C. Goble A Fresh Look at FAIR for Research Software. arXiv (2021). 2101.10883. https://arxiv.org/abs/2101.10883

Mölder F, Jablonski KP, Letcher B, Hall M, Tomkins-Tinch C, **Sochat V**, et al. Sustainable data analysis with Snakemake [version 1; peer review: awaiting peer review]. F1000Research 2021, 10:33 (https://doi.org/10.12688/f1000research.29032.1)

Sochat, V. (2020). AskCI Server: Collaborative knowledge base. In Practice and Experience in Advanced Research Computing (PEARC ’20). Association for Computing Machinery, New York, NY, USA, 514–517. DOI:https://doi.org/10.1145/3311790.3399616

Sochat, V., (2020). GridTest: testing and metrics collection for Python. Journal of Open Source Software, 5(51), 2284, https://doi.org/10.21105/joss.02284

Sochat, V (2019). Singularity Compose: Orchestration for Singularity Instances. Journal of Open Source Software, 4(40), 1578, https://doi.org/10.21105/joss.01578

Sochat, V (2019). Container Tree: Software to Model Container Filesystems, Packages, and Inheritance. Journal of Open Source Software, 4(37), 1418, https://doi.org/10.21105/joss.01418

Sochat, V (2019). WatchMe: Software for Reproducible Monitoring and Data Collection. Journal of Open Source Software, 4(37), 1388, https://doi.org/10.21105/joss.01388

Sochat, V (2018). Containershare: Open Source Registry to build, test, deploy with CircleCI . Journal of Open Source Software, 3(28), 878, https://doi.org/10.21105/joss.00878

Sochat, V (2018). HelpMe Command Line Helper Utility . Journal of Open Source Software, 3(26), 775, https://doi.org/10.21105/joss.00775

Sochat V, (2018). The Scientific Filesystem. GigaScience, giy023, https://doi.org/10.1093/gigascience/giy023

Sochat V, (2018). The Experiment Factory: Reproducible Experiment Containers. Journal of Open Source Software, 3(22), 521, https://doi.org/10.21105/joss.00521

Sochat V, Prybol CJ, Kurtzer GM (2017) Enhancing reproducibility in scientific computing: Metrics and registry for Singularity containers. PLoS ONE 12(11): e0188511. https://doi.org/10.1371/journal.pone.0188511

Sochat V, (2017), Singularity Registry: Open Source Registry for Singularity Images, Journal of Open Source Software, 2(18), 426, doi:10.21105/joss.00426

Kurtzer GM, **Sochat V**, Bauer MW (2017) Singularity: Scientific containers for mobility of compute. PLoS ONE 12(5): e0177459.

Sochat V, Eisenberg IW, Enkavi AZ, Li J, Bissett PG and Poldrack RA. The Experiment Factory: standardizing behavioral experiments. Front. Psychol. 2016.

Durnez J, Degryse J, Moerkerke B, Seurinck R, **Sochat V**, Poldrack R, Nichols T. Power and sample size calculations for fMRI studies based on the prevalence of active peaks. bioRxiv, 2016.

Sochat V, Gorgolewski KJ, Koyejo O, Durnez J, Poldrack RA. Effects of thresholding on correlation-based image similarity metrics. Frontiers in Neuroscience. 2015.

Sochat V, AuthorSynth: a collaboration network and behaviorally-based visualization tool of activation reports from the neuroscience literature. Frontiers in Neuroinformatics. 2015.

Poldrack, R, Laumann T, Koyejo O, Gregory B, Hover A, Chen MY, Gorgolewski KJ, Luci J, Joo SJ, Boyd R, Hunicke-Smith S, Simpson Z, Caven T, **Sochat V**, Shine J, et al. "Long-Term Neural, Behavioral, and Physiological Phenotyping of a Single Human: The MyConnectome Project" Nature Communications. 2015.

Sochat V, Supekar K, Bustillo J, Calhoun V, Turner JA, et al. A Robust Classifier to Distinguish Noise from

fMRI Independent Components. PLoS ONE. 2014.

S. Finlayson, **V. Sochat**, L. Szabo, L. Yancy Jr. A Rapid Learning System for Personalized Glioblastoma Treatment Planning. Abstract presentation at the AMIA Annual Symposium, Washington DC, USA. 2013.

TALKS AND PRESENTATIONS

Vanessa Sochat. "Converged Computing and Agentic Workflows" Main track talk, HPSF 2026.

Vanessa Socha. "Kubeflow Trainer: Flux for HPC Support." Virtual tutorial, KubeCon EU March, 2026.

Vanessa Sochat and Daniel Milroy. "The Future of Discovery: Converged Computing, Agentic Science, and the Evolution of the HPC Center." Invited talk, hybrid HPC cloud conference (CNRS) Paris, France.

Vanessa Sochat, Evan Bollig, Tim Brown, J. Lowell Wofford, Loic Pottier, Daniel Milroy. Orchestrating Complex HPC and AI/ML Workflows on Kubernetes Using Flux and AWS. Supercomputing 2025 Tutorial.

V. Sochat, D. Milroy 2025. “Flux: Next-Generation Workload Management for High Performance Computing and Cloud.” HPCKP. July 1, 2025.

Sochat V (2024, May 8). "Compatibility for HPC, a Story of Rainbows and Schedulers ," HPC Knowledge Meeting 2024. https://hpckp.org/talks/compatibility-for-hpc-a-story-of-rainbows-and-schedulers/

Sochat V (2024, May 15). "Containers in HPC: Community Survey 2024 ," Annual High Performance Container Workshop. https://container-in-hpc.org/isc/2024/hpcw/index.html.

Sochat V, Milroy D, Fox D, (2024, February). "Kubernetes and HPC: Bare Metal Bros ," FOSDEM HPC, Big Data, and Data Science Devroom, Brussels, Belgium.
https://fosdem.org/2024/schedule/event/fosdem-2024-2590-kubernetes-and-hpc-bare-metal-bros/

Gharaibeh, A, **Sochat V,** (2023, October). "On-Demand Systems and Scaled Training Using the JobSet API ," Kubecon America, Chicago, 2023.

Sochat V (2023, September). "Open Source Introspection," 2023 UC Open Source Symposium Keynote

Sochat V, Misale, C (2023, May). "Cloud and HPC Convergence: Flux for Job Management on Kubernetes", HPC Knowledge Meeting 2023
https://hpckp.org/talks/cloud-and-hpc-convergence-flux-for-job-management-on-kubernetes/

Sochat, V, Sill, Alan (2023, April 24) “The hpc.social project" EasyBuild users meeting.
https://www.youtube.com/live/lysx3e8pDPs

Sochat, V, Woźniak, Michał (2023, April 21) “Enabling HPC and ML Workloads with the Latest Kubernetes Job Features." Accepted talk for Kubecon, Amsterdam.

Invited Speaker for UC Boulder CS Colloquium (2022, 7 April). https://youtu.be/2Oet0hGOy7U
https://calendar.colorado.edu/event/cs_colloquium_vanessa_sochat_on_the_software_complexity_puzzle

Sochat, V (2022, April 8) “The Software Complexity Puzzle” Invited Speaker for UC Boulder CS Colloquium. https://calendar.colorado.edu/event/cs_colloquium_vanessa_sochat_on_the_software_complexity_puzzle

Sochat, V (2022, March 7) “The Research Software Engineer Movement (repeat)” Open Planetary Foundation.

Sochat, V (2022, January) “The Research Software Engineer Movement” Speaker for EasyBuild Users Meeting.

Sochat, V (2021, September) “The Stories We Tell Ourselves” Keynote for SeptembRSE, International Research Software Engineering Conference.

Sochat, V (2020, September) “Singularity Container on Google Cloud Platform (GCP) Tutorial” Co-host for Google HPC Days, https://cloudonair.withgoogle.com/events/singularity_containters_on_gcp_tutorial.

Sochat, V (2020, September) “Extensions to Schema.org for structured, semantic & executable research documents” Invited panelist, Stencila Community Webinar https://stenci.la/blog/2020-09-20-register-now-for-stencila-community-call-thur-24-sept-2020/.

Sochat, V (2020, September) “Research Software Directories, What, Why, and How?” Series of Online Research Software Events (SORSE) 2020 Talks and Discussion, https://sorse.github.io/programme/discussions/event-013/.

Sochat, V (2020, July) “The Research Software Encyclopedia” Invited talk, Research Software Engineers Community Workshop, 2020, PEARC20, https://us-rse.org/events/2020/2020-07-pearc20.

Sochat, V (2020, July) “The Singularity Executor: A Contributor Overflow Exception” Invited talk, Apache Airflow Summit 2020, https://airflowsummit.org/speakers/vanessa-sochat/.

Sochat V, (2020, June) “Singularity Containers” Invited talk, Dataverse Community Meeting, https://projects.iq.harvard.edu/dcm2020/people/vanessa-sochat.

Sochat V, (2020, January 10) “Research Software Engineering: A Future at Stanford” Keynote: Campus IT Plan Research Summit, Stanford CA, https://itcommunity.stanford.edu/events/campus-it-plan-research-summit.

Sochat V, (2019, December 19) “Research Software Engineers: A New Career” Lightning Talk: IT Unconference, Stanford CA, https://itcommunity.stanford.edu/unconference.

Sochat V, (2018, April 4) “The Scientific Filesystem” Invited Speaker: Containers in HPC Symposium at UCAR, Boulder CO, https://sea.ucar.edu/conference/2018/containers.

Sochat V, (2018, March 7) “Introduction to Singularity” Invited Speaker: CyVerse Container Camp: Container Technology for Scientific Research, University of Arizona, Tuscon AZ.

Sochat V, (2017, July 11). “Reproducibility and Containers: The Perfect Sandwich” Invited Speaker: Practice & Experience in Advanced Research Computing, New Orleans LA. https://www.pearc17.pearc.org/speakers

Sochat V, (2017, February). “Singularity Containers for Scientific Compute” Talk Stanford Genomics Cluster User Group, Stanford CA, USA.

Sochat V, (2015, October). “Building Tools for Neuroimaging: the intersection of high performance computing, web technology, and fun in graduate school.”, Talk for Research Computing Group, Stanford CA, USA.

Sochat V, (2015, March). “Brain Maps Like Mine content-aware image comparison and retrieval for interactive visualization and meta-analysis of brain statistical maps”, Research in Progress Talk, Stanford CA, USA.

Sochat V, (2014, June). “Introduction to Machine Learning,” SIMR Summer Research Program, Stanford CA, USA.

Sochat V, (2014, May). “Machine Learning for Images,” Biomedical Imaging Analysis & Interpretation Lecture, Stanford CA, USA.

Sochat V, (2013, May). “Neuroinformatics,” Biomedical Imaging Analysis and Interpretation Lecture, Stanford CA, USA.